2. 決定一個網格的量值

(1) 取座標 $(y,z)=(0,-13.01)$ 的節點說明

(2) 將問題簡化成 2D, 計算整個三角形面積, 定義向量:

$$\begin{cases}\vec{V}_1 = P_2 - P_1 = (-3.51\times10^{-3},\ -7.77\times10^{-3})\\ \vec{V}_2 = P_3 - P_1 = (-7\times10^{-5},\ -8.02\times10^{-3})\end{cases}$$

整個三角形面積:

$$A_{tot} = \frac{1}{2}\left|\vec{V}_1 \times \vec{V}_2\right| = 1.38\times10^{-5}\,\mathrm{mm}^2$$

子三角形面積與權重:

$$\begin{aligned} A_1 &= (Q,P_2,P_3) = 1.177\times10^{-5}\,\mathrm{mm}^2\\ A_2 &= (Q,P_1,P_3) = 0.0255\times10^{-5}\,\mathrm{mm}^2\\ A_3 &= (Q,P_2,P_1) = 0.17661\times10^{-5}\,\mathrm{mm}^2\end{aligned} \quad\Rightarrow\quad \begin{cases} W_1 = \dfrac{A_1}{A_{tot}} = \dfrac{1.177\times10^{-5}}{1.38\times10^{-5}} = 0.853\\ W_2 = \dfrac{A_2}{A_{tot}} = 0.0185\\ W_3 = \dfrac{A_3}{A_{tot}} = 0.128\end{cases}$$

(3) 由重心插值得磁通密度與磁通量

因此

$$B(0,-13.01) = B_{x_1}\cdot W_1 + B_{x_2}\cdot W_2 + B_{x_3}\cdot W_3 = 257.7\,\mathrm{mT}$$

故

$$\Delta\Phi = B(0,-13.01)\cdot\Delta A = 257.7\times10^{-3}\cdot 0.5\times10^{-12} = 128.85\,\mathrm{fWb}$$

(4) 實際使用 3D 插值

$$B(0,-13.01) = 362.733\,\mathrm{mT}$$

故

$$\Delta\Phi = 362.733\,\mathrm{mT}\cdot 0.5\,\mu\mathrm{m}^2 = 181.4\,\mathrm{fWb}$$

全部加總:

$$\boxed{\Phi = \sum \Delta\Phi = 0.572\,\mathrm{nWb}}$$